Pulsar API

A lightweight REST API for real-time pub/sub messaging.
Self-hostable, zero vendor lock-in, WebSocket support included.

Base URL

```
$
https://api.pulsar.example.com/v1
```

All requests must include:

```
$
Content-Type: application/json
Authorization: Bearer <your_api_key>
```

Endpoints

Topics

Method	Endpoint	Description
GET	`/topics`	List all topics
POST	`/topics`	Create a topic
DELETE	`/topics/:id`	Delete a topic
GET	`/topics/:id/messages`	Fetch recent messages

Subscriptions

Method	Endpoint	Description
GET	`/subscriptions`	List subscriptions
POST	`/subscriptions`	Create a subscription
DELETE	`/subscriptions/:id`	Cancel a subscription

Create a Topic

POST `/topics`

Request body:

```
$
{
  "name": "order-events",
  "retention_hours": 48,
  "max_message_size_kb": 256
}
```

Response:

```
$
{
  "id": "top_a8f3k2",
  "name": "order-events",
  "created_at": "2024-11-01T10:22:00Z",
  "retention_hours": 48,
  "max_message_size_kb": 256
}
```

Publish a Message

POST `/topics/:id/publish`

```
$
{
  "payload": {
    "order_id": "ord_991",
    "status": "shipped",
    "timestamp": "2024-11-01T10:30:00Z"
  },
  "key": "order-991"
}
```

Client Examples

Python

```python
$
import httpx

client = httpx.Client(
    base_url="https://api.pulsar.example.com/v1",
    headers={"Authorization": "Bearer sk_live_xxxx"},
)

# Publish
client.post("/topics/top_a8f3k2/publish", json={
    "payload": {"event": "user.signup", "user_id": 42},
    "key": "user-42",
})

# Fetch recent messages
resp = client.get("/topics/top_a8f3k2/messages", params={"limit": 10})
messages = resp.json()["messages"]
```

JavaScript

```javascript
$
const res = await fetch("https://api.pulsar.example.com/v1/topics/top_a8f3k2/publish", {
  method: "POST",
  headers: {
    "Authorization": "Bearer sk_live_xxxx",
    "Content-Type": "application/json",
  },
  body: JSON.stringify({
    payload: { event: "page.view", path: "/home" },
    key: "session-123",
  }),
});

const data = await res.json();
console.log(data.message_id);
```

Go

```
$
package main

import (
    "bytes"
    "encoding/json"
    "fmt"
    "net/http"
)

func publish(topicID string, payload any) error {
    body, _ := json.Marshal(map[string]any{"payload": payload})
    req, _ := http.NewRequest("POST",
        fmt.Sprintf("https://api.pulsar.example.com/v1/topics/%s/publish", topicID),
        bytes.NewReader(body),
    )
    req.Header.Set("Authorization", "Bearer sk_live_xxxx")
    req.Header.Set("Content-Type", "application/json")
    _, err := http.DefaultClient.Do(req)
    return err
}
```

WebSocket Streaming

Connect to receive messages in real time:

```
$
wss://api.pulsar.example.com/v1/topics/:id/stream?token=<api_key>
```

```
$
const ws = new WebSocket(
  "wss://api.pulsar.example.com/v1/topics/top_a8f3k2/stream?token=sk_live_xxxx"
);

ws.onmessage = (event) => {
  const msg = JSON.parse(event.data);
  console.log("received:", msg.payload);
};
```

Error Codes

Code	Meaning
400	Bad request / malformed JSON
401	Invalid or missing API key
403	Insufficient permissions
404	Topic or resource not found
413	Payload exceeds max size
429	Rate limit exceeded
500	Internal server error

Rate Limits

Plan	Requests/min	Messages/day
Free	60	10,000
Pro	600	1,000,000
Business	6,000	Unlimited

Self Hosting

```
$
docker run -d \
  -p 8080:8080 \
  -e SECRET_KEY=your_secret \
  -e DATABASE_URL=postgres://... \
  ghcr.io/example/pulsar:latest
```

License

Apache 2.0